Kshitij Sinha

Mathematics
Indian Institute of Technology Bombay

23D0781
Ph.D
Male

Examination	University	Institute	Year	CPI/%
Ph.D	IIT Bombay	IIT Bombay	2023-Present	9.48
M.Sc.	IIT Kanpur	IIT Kanpur	2020-22	7.50
B.Sc.	BHU	BHU, Varanasi	2017-20	8.26
Intermediate/+2	CBSE	Udai Pratap Public School	2015-17	91.80
Matriculation	ICSE	St.John's School	2015	82.80

EXAMS QUALIFIED

- **BHU-UET** *(2017)*
- **IIT JAM** (All India rank of 77) *(2020)*
- **GATE** (All India rank of 30) *(2023)*
- **NBHM** *(2023)*
- **CSIR-NET**(All India rank of 89) *(2023)*

PUBLICATIONS

- **Kshitij Sinha**. Quantitative periodic homogenization of parabolic equations with large drift and potential (arXiv.2509.22003)
- **Saumyajit Das, Kshitij Sinha**. Homogenization of Three Species Reaction Diffusion Equation in Perforated Domains (arXiv:2603.26927v1).

SEMINARS, CONFERENCES ATTENDED

- **NCMW - A Contemporary Course on Elliptic Partial Differential Equations** *(20 May '24 - 01 Jun'24)*
  - Organized by Professor Ali Hyder(TIFR-CAM) and Professor Debabrata Karmakar(TIFR-CAM) at TIFR - Center for Applicable Mathematics(CAM), Bengaluru
- **Quantitative theory of Homogenization** *(17 Feb '25 - 22 Feb'25)*
  - Organized by Professor Harsha Hutridurga(IIT-B) and Professor S. Sivaji Ganesh(IIT-B) at IIT Bombay
- **Stochastic Processes and Related Topics 2025** *(13 Sep '25 - 15 Sep'25)*
  - Organized by Professor Chetan D. Pahlajani(IIT-Gandhinagar) at IIT Gandhinagar

TALKS GIVEN

- **Quantitative theory of Homogenization** *(17 Feb '25 - 22 Feb'25)*
  - Doctoral Course(5:30 hours) on the Quantitative aspects of Homogenization of Elliptic Partial Differential equations in periodic settings
- **Stochastic Processes and Related Topics 2025** *(13 Sep '25 - 15 Sep'25)*
  - Talk on Quantitative periodic homogenization of parabolic equations with large drift and potential

TEACHING EXPERIENCE

- **Teaching Assistant** for the course **MA515 : Partial Differential Equations**, Tutorials and Grading *(Aug'24-Dec'24 & Aug'25 - Dec'25)*
- **Admin Teaching Assistant** for the course **SI416 : Optimization**, Tutorials and Grading *(Jan'25-Apr'25 & Jan'26-Apr'26)*

TECHNICAL SKILLS

- **Programming & Scripting Languages**: C, C++
- **Tools & Libraries**: MATLAB, LaTeX

EXTRA-CURRICULARS

- **Cultural Secretary** at Hostel 14, IIT Bombay *(Aug'25-Aug'26)*
- **Ram-Leela** played the roles of Kumbhakaran and Banasur in the Ram-Leela
- **Anime** suggestions: Assassination Classroom, Demon Slayer

LANGUAGES

- **Hindi** Mother tongue
- **English** Good